

Electromagnetism & Electrodynamics HW4

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Technion

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1 Scalar Field

A scalar field $\phi(x)$ is given, therefore under lorentz transformations it is invariant. We'll develop the equations of motion of a free real field given by the following Lagrangian density

$$\mathcal{L} = \frac{1}{2} \partial^\mu \phi \partial_\mu \phi \quad (1)$$

1. Derive the equations of motion. What equation is this?
2. We'll add another element to the Lagrangian

$$\mathcal{L} = \frac{1}{2} \partial^\mu \phi \partial_\mu \phi - \frac{1}{2} m^2 \phi^2 - \frac{\lambda}{4} \phi^4 \quad (2)$$

find the new equations of motion.

1.1 Equations of Motion

Using the EL equations

$$\frac{\partial \mathcal{L}}{\partial \phi} = 0 \quad (3)$$

$$\frac{\partial \mathcal{L}}{\partial \partial_\nu \phi} = \frac{1}{2} \left(\frac{\partial \partial^\mu \phi}{\partial \partial_\nu \phi} \partial_\mu \phi + \frac{\partial \partial_\mu \phi}{\partial \partial_\nu \phi} \partial^\mu \phi \right) \quad (4)$$

$$= \frac{1}{2} (g^{\mu\nu} \delta_\sigma^\nu \partial_\mu \phi + \delta_\mu^\nu \partial^\mu \phi) \quad (5)$$

$$= \partial^\nu \phi \quad (6)$$

therefore the equations of motion are

$$\square \phi = 0 \quad (7)$$

1.2 Another Element

The only change to the EL equations is to the element depending on the field itself

$$\frac{\partial \mathcal{L}}{\partial \phi} = -m^2 \phi - \lambda \phi^3 \quad (8)$$

giving us

$$\square \phi + m^2 \phi + \lambda \phi^3 = 0 \quad (9)$$

2 Long Lagrangian

The following Lagrangian is given

$$\mathcal{L} = \mathcal{L}_{EM} + \mathcal{L}_k + \mathcal{L}_{int} = -\frac{1}{16\pi c} F_{\mu\nu} F^{\mu\nu} + \frac{k^2}{8\pi c} A_\mu A^\mu - \frac{1}{c^2} J_\mu A^\mu \quad (10)$$

1. What are the units of k and ck ?
2. Is the new action A lorentz scalar?
3. Is it invariant for gauge transformations?
4. Find the equations of motion
5. Apply divergence to both sides and show the continuity equation does not hold unless $\partial_\mu A^\mu = 0$
6. Using the last part show that the equations of motion can be written as

$$(\square + k^2) A^\mu = \frac{4\pi}{c} J^\mu \quad (11)$$

7. Show that in the static limit for a point charge q at the origin the solution is

$$\phi(r) = \frac{q}{r} e^{-kr} \quad (12)$$

2.1 Units

Since all elements must have the same units we can demand that

$$\left[-\frac{1}{16\pi c} F_{\mu\nu} F^{\mu\nu} \right] = \left[\frac{k^2}{8\pi c} A_\mu A^\mu \right] \quad (13)$$

we know that

$$[F_{\mu\nu}] = [\partial_\mu A_\nu] = \frac{[A]}{L} \quad (14)$$

so

$$\frac{[A]^2}{L^2} = [k]^2 [A]^2 \implies [k] = \frac{1}{L}, \quad [ck] = \frac{1}{T} \quad (15)$$

2.2 Lorentz Scalar

$F_{\mu\nu} F^{\mu\nu}$ is known to be a lorentz scalar so we continue to the other elements.

$$A_\mu A^\mu \rightarrow (\Lambda_\mu{}^\nu A_\nu) (\Lambda^\mu{}_\sigma A^\sigma) = \Lambda_\mu{}^\nu \Lambda^\mu{}_\sigma A_\nu A^\sigma = \delta_\sigma^\nu A_\nu A^\sigma = A^\nu A_\nu \quad (16)$$

therefore this element is a lorentz scalar. As for the last element the process is identical to this entirely just with the name J instead of one of the As so I won't rewrite it. In total, the action is a lorentz scalar.

2.3 Gauge Invariance

A gauge transformation is

$$A_\mu \rightarrow A_\mu + \partial_\mu \Lambda \quad (17)$$

where Λ is a function of the coordinates. Looking at the kinetic element

$$A_\mu A^\mu \rightarrow (A_\mu + \partial_\mu \Lambda) (A^\mu + \partial^\mu \Lambda) \quad (18)$$

$$= A_\mu A^\mu + A_\mu \partial^\mu \Lambda + A^\mu \partial_\mu \Lambda + \partial_\mu \Lambda \partial^\mu \Lambda \quad (19)$$

which does not automatically always equal $A_\mu A^\mu$, nor is it a total derivative, so this is not gauge invariant, making the entire Lagrangian not gauge invariant.

2.4 Equations of Motion

Looking at a subset of the lagrangian, $\mathcal{L}_{int} + \mathcal{L}_{EM}$ we have a known problem we solved in the lecture whose equation of motion are

$$\partial^\nu F_{\nu\mu} = \frac{4\pi}{c} J_\mu \quad (20)$$

therefore we only need to solve for the kinetic element

$$\frac{\partial \mathcal{L}_k}{\partial A^\nu} = \frac{k^2}{8\pi c} \left(\frac{\partial A_\mu}{\partial A^\nu} A^\mu + \frac{\partial A^\mu}{\partial A^\nu} A_\mu \right) \quad (21)$$

$$= \frac{k^2}{8\pi c} \left(\frac{\partial A^\sigma}{\partial A^\nu} g_{\mu\sigma} A^\mu + A_\mu \delta^\mu_\nu \right) \quad (22)$$

$$= \frac{k^2}{8\pi c} (\delta^\sigma_\nu g_{\mu\sigma} A^\mu + A_\mu \delta^\mu_\nu) \quad (23)$$

$$= \frac{k^2}{4\pi c} A_\nu \quad (24)$$

and

$$\frac{\partial \mathcal{L}_k}{\partial \partial_n u A_\mu} = 0 \quad (25)$$

so the equations of motion are

$$\partial^\nu F_{\nu\mu} + k^2 A_\mu = \frac{4\pi}{c} J_\mu \quad (26)$$

2.5 Divergence

directly

$$\partial^\mu \partial^\nu F_{\nu\mu} + \partial^\mu k^2 A_\mu = \partial^\mu \frac{4\pi}{c} J_\mu \quad (27)$$

but since $\partial^\mu \partial^\nu$ is symmetric and $F_{\mu\nu}$ is antisymmetric this element is zero.

$$\partial^\mu k^2 A_\mu = \partial^\mu \frac{4\pi}{c} J_\mu \quad (28)$$

and we can directly see that for the continuity equation to hold, we must have $\partial^\mu A_\mu = 0$.

2.6 Box

returning to the equation of motion and opening terms in it

$$\partial^\nu F_{\nu\mu} + k^2 A_\mu = \frac{4\pi}{c} J_\mu \quad (29)$$

$$\partial^\nu (\partial_\nu A_\mu - \partial_\mu A_\nu) + k^2 A_\mu = \frac{4\pi}{c} J_\mu \quad (30)$$

$$(\partial_\nu \partial^\nu A_\mu - \partial_\mu \partial^\nu A_\nu) + k^2 A_\mu = \frac{4\pi}{c} J_\mu \quad (31)$$

$$(\square A_\mu - 0) + k^2 A_\mu = \frac{4\pi}{c} J_\mu \quad (32)$$

$$(\square + k^2) A_\mu = \frac{4\pi}{c} J_\mu \quad (33)$$

2.7 Static limit

Since we are in the static limit, the 4-current is of the form

$$j^\mu = (c\rho, \mathbf{0}) \quad (34)$$

where

$$\rho = q\delta^3(r) \quad (35)$$

inserting into the last part and looking at the time component

$$(-\nabla^2 + k^2) \phi = 4\pi q\delta^3(r) \quad (36)$$

remembering we are in spherical coordinates so the laplacian operator is

$$\nabla^2 \phi = \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d\phi}{dr} \right) \quad (37)$$

finding the derivatives

$$\frac{d}{dr} \left(\frac{q}{r} e^{-kr} \right) = q \left(-\frac{1}{r^2} e^{-kr} - \frac{k}{r} e^{-kr} \right) = -q e^{-kr} \left(\frac{1}{r^2} + \frac{k}{r} \right) \quad (38)$$

$$\frac{d}{dr} (-q e^{-kr} (1 + kr)) = -q (-k e^{-kr} (1 + kr) + k e^{-kr}) = q k^2 r e^{-kr} \quad (39)$$

therefore

$$\nabla^2 \phi = \frac{q k^2}{r} e^{-kr} = k^2 \phi \quad (40)$$

and when $r \neq 0$ we get

$$0 = (k^2 - k^2) \phi = 4\pi q\delta^3(r) = 0 \quad (41)$$

The only point left to check is the origin. Since near the origin $e^{-kr} \rightarrow 1$, the potential behaves like

$$\phi \sim \frac{q}{r}.$$

To see the coefficient of the source, we integrate over a small sphere of radius ε around the origin:

$$\int_V (-\nabla^2 + k^2)\phi d^3x.$$

The $k^2\phi$ term goes to zero when $\varepsilon \rightarrow 0$, because $\phi \sim q/r$ (so after it's multiplied by the jacobian it goes like r making the integral of that element tend to 0). Therefore only the Laplacian term contributes:

$$\int_V (-\nabla^2\phi)d^3x = - \oint_{\partial V} \nabla\phi \cdot d\mathbf{S}.$$

Near the origin

$$\frac{d\phi}{dr} \sim -\frac{q}{r^2},$$

so

$$- \oint_{\partial V} \nabla\phi \cdot d\mathbf{S} = - \left(-\frac{q}{\varepsilon^2}\right) 4\pi\varepsilon^2 = 4\pi q.$$

So the expression is zero everywhere except the origin, and gives total weight $4\pi q$ at the origin. Therefore

$$(-\nabla^2 + k^2)\phi = 4\pi q\delta^3(r).$$

Since for a static point charge

$$J^0 = cq\delta^3(r),$$

we get

$$\frac{4\pi}{c}J^0 = 4\pi q\delta^3(r),$$

so the solution is correct at the origin too.